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Atlantic Technological University

MSSIA

Multi-Sensor Spatial Intelligence & Analytics Platform

Real-time indoor environment monitoring via ESP32 sensors, FastAPI, and an interactive floor plan heatmap dashboard

GitHub: github.com/Nokuvimba/Final-Year-Project

Live site: final-year-project-alpha-azure.vercel.app

Scan to visit live MSSIA site

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Project Summary

The Problem

While studying at ATU Galway, I noticed that certain areas of the campus had noticeably weak Wi-Fi coverage. Students and staff had no way to visualize where signal dropped or why. At the same time, room conditions like temperature, humidity and air quality went unmonitored in most rooms.

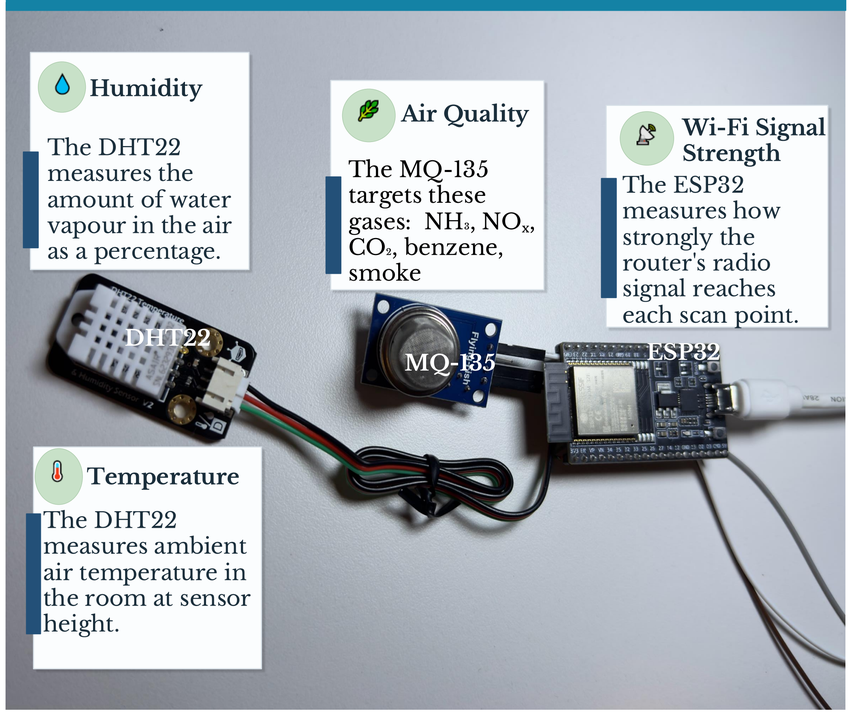
The Idea

Why not build a spatial data visualization system that gives facilities teams the tools to identify and fix these problems — while also letting students and staff see the environment around them in real time? MSSIA was born from that question.

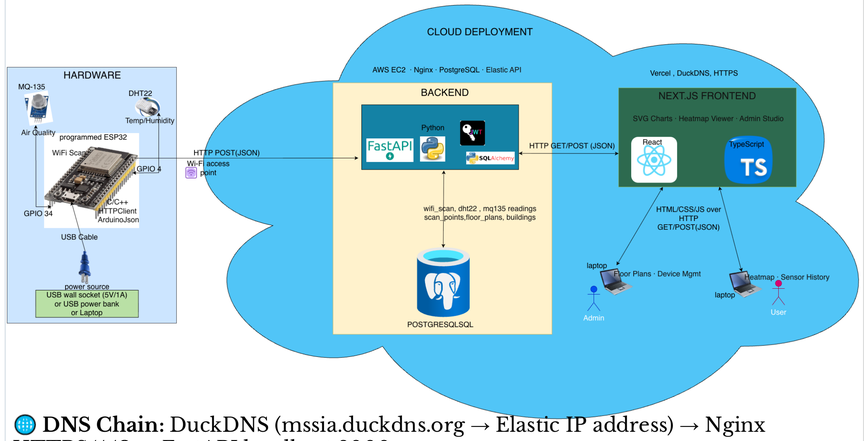
What MSSIA Does

It ingests numeric readings from any sensor-equipped device and overlays them as live interactive heatmaps on uploaded floor plans. This project demonstrates the concept using four sensor types: Wi-Fi signal strength, temperature, humidity and air quality. Any measurable value from any sensor can be added.

Hardware

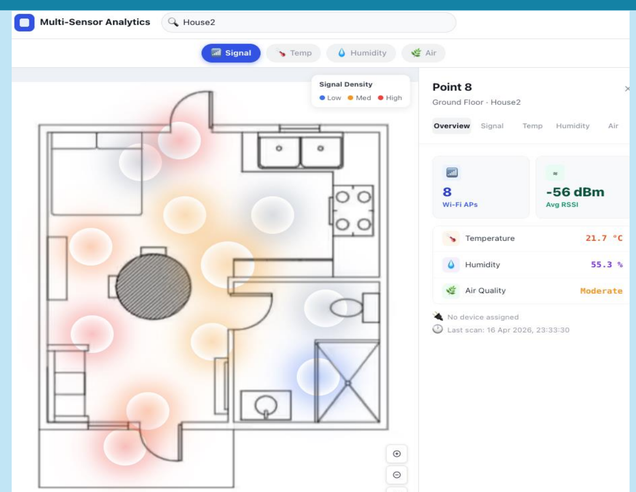


System Architecture



🌐 **DNS Chain:** DuckDNS (mssia.duckdns.org → Elastic IP address) → Nginx HTTPS/443 → FastAPI localhost:8000
Vercel frontend calls mssia.duckdns.org directly from the user's browser via HTTPS (Let's Encrypt certificate).

MSSIA Website



Results

Admin → System → User Flow Diagram

Software & Technologies

Backend (Python)

- FastAPI
- JWT authorization

Database

- PostgreSQL - relational database
- SQLAlchemy

Frontend (TypeScript)

- Next.js 16
- React 18
- Custom SVG line charts (no chart library)
- JWT auth guard – local Storage token

Hardware / Embedded (C/C++)

- Arduino framework
- WiFi.h · HTTPClient.h · ArduinoJson

Cloud & DevOps

- AWS EC2 t2.micro - FastAPI + PostgreSQL + Nginx, Elastic IP
- Vercel - Next.js hosting (auto-deploy from GitHub)
- DuckDNS + Let's Encrypt - free HTTPS (mssia.duckdns.org)

Conclusion

Did It Work?

Yes — MSSIA proves that a single generic platform can ingest readings from any sensor, map them spatially onto a floor plan, and visualize them as live interactive heatmaps accessible from any browser. The four sensor types demonstrated (Wi-Fi, temperature, humidity, air quality) each produce distinct, colour-differentiated heatmaps — confirming the concept works end-to-end from hardware to cloud to user.

✅ Extensible: new sensors can still be added with minor code changes needed